

Guideline

Head injury in children

1 Scope

For the treatment of children within the Trust.

2 Purpose

To help the clinician dealing with a child presenting after a head injury in accordance with NICE guidelines and taking local provisions into consideration.

3 Definitions

3.1 Head injury

'Head injury' for the purposes of the guideline is defined as any trauma to the head, other than superficial injuries to the face. Please refer to NICE clinical guideline 176.

3.2 Abbreviations used

A&E	accident and emergency/ emergency department
ABC	airway, breathing, circulation (resuscitation mnemonic)
ATLS	advanced trauma and life support
APLS	advanced paediatric life support
COU	children's observation unit
CT	computed tomography
GCS	Glasgow coma scale
ITP	idiopathic thrombocytopaenia
NICE	National Institute for Health and Care Excellence
PICU	paediatric intensive care unit

4 Introduction

The primary patient outcome of concern throughout the guideline is clinically important brain injury. In the UK there are more than one million attendances for head injury per year, children under 16 years of age making up 50% of these.

The risk for intracranial complications is 1-3%; the challenge is to recognise early signs of significant head injury and manage them appropriately.

5 History and examination

Significant features of history and examination need to be seen in context as there is no predictor or combination of predictors that is 100% sensitive for detecting intracranial injury and it is not possible to address every possible clinical situation. The need to rule out significant intracranial injury needs to be weighed against the exposure to radiation and potential side effects of sedation for imaging which may be required for small children.

There should be a very low threshold to discuss any case with an emergency medicine consultant. If it is anticipated that the child may require observation and investigations and is neurologically stable, early admission to F3 should be arranged.

5.1 History

Pertinent features in the history are the time and mechanism of injury, any loss of consciousness or drowsiness, seizure activity, any episodes of vomiting, amnesia, blood or fluid leak from the ears/ nose.

Safeguarding – consider or suspect abuse as a contributory factor to or cause of head injury in children.

5.2 Examination

The following features on examination are important with regards to the further management and their presence or absence must be recorded in the patient's notes.

Feature on examination	Comment/ recommendation
Signs of basal skull fracture • Haemotympanum • 'Panda' eyes • Cerebrospinal fluid leak • Battle's sign	CT scan of the head within one hour.
Open or depressed skull fracture	
Focal neurological deficit	
GCS equal to or less than 13	
Tense fontanelle in an infant	CT scan within one hour. Requires careful assessment for further injuries and the possibility of inflicted injury.

6 Management

6.1 Severe head injury

The priority for all patients is the stabilisation of airway with cervical spine control, breathing and circulation (ABC) before attention to other injuries. Please refer to the relevant ATLS/APLS manual for detailed management guidelines.

Please refer to the Trauma East Manual of Procedures and Operations for the management of severe head injury.

If indicated, children aged ten years or more can be treated as adults for the purposes of cervical spine imaging. Children under ten years should receive anterior/ posterior and lateral views without an anterior/ posterior peg view.

6.2 Patient monitoring

For patients admitted for head injury observation the minimum acceptable documented neurological observations are:

- GCS
- limb movements
- heart rate
- temperature
- pupil size and reactivity
- respiratory rate
- blood pressure
- blood oxygen saturation

Observations should be performed and recorded on a half-hourly basis until GCS equal to 15 has been achieved. The minimum frequency of observations for patients should be as follows, starting after the initial assessment in A&E:

- half-hourly for two hours
- then one-hourly for four hours
- then two-hourly thereafter

Period of observation: Most significant complications after head injury occur within 24 hours. The minimum period of observation should be four to six hours.

Infants and young children should be observed on COU or be admitted as inpatients to the wards, as staff in these areas have the appropriate experience.

Any of the following examples of neurological deterioration should prompt **urgent re-assessment** by the supervising doctor:

- development of agitation or abnormal behaviour
- a sustained (for >30 minutes) drop of one point in GCS level (greater weight should be given to a drop of one point in the motor score of the GCS)

- any drop of greater than two points in GCS level regardless of duration or GCS sub-scale
- development of severe or increasing headache or persisting vomiting
- new or evolving neurological symptoms or signs such as pupil inequality or asymmetry of limb or facial movement

6.2.1 Modified Glasgow coma scale for children

Best eye response (4) 1. No eye opening 2. Eye opening to pain 3. Eye opening to verbal command 4. Eyes open spontaneously	
Best verbal response (5) 1. No vocal response 2. Occasionally whimpers and/or moans 3. Cries inappropriately 4. Less than usual ability and/or spontaneous irritable cry 5. Alert, babbles, coos, words or sentences to usual ability	Best grimace response (pre-verbal age) (5) 1. No response to pain 2. Mild grimace to pain 3. Vigorous grimace to pain 4. Less than usual spontaneous ability or only response to touch stimuli 5. Spontaneous normal facial/ oro-motor activity
Best motor response (6) 1. No motor response to pain 2. Abnormal extension to pain (decerebrate) 3. Abnormal flexion to pain (decorticate) 4. Withdrawal to painful stimuli 5. Localises to painful stimuli or withdraws to touch 6. Obeys commands or performs normal spontaneous movements	

6.3 Criteria for admission to inpatient ward

- Patients with new, clinically significant abnormalities on imaging.
- Patients who have not returned to GCS equal to 15 after imaging, regardless of the imaging results.
- When a patient fulfils the criteria for CT scanning but this cannot be done within the appropriate period, either because CT is not available or because the patient is not sufficiently co-operative to allow scanning.
- Continuing worrying signs of concern to the clinician, eg:
 - persistent vomiting
 - severe headaches

- Other sources of concern to the clinician, eg:
 - drug or alcohol intoxication
 - other injuries
 - shock
 - suspected inflicted injury
 - meningism
 - cerebrospinal fluid leak
 - social concerns

6.4 Criteria for performing CT head

6.4.1 For children who have sustained a head injury and have any of the following risk factors, perform a CT head scan within one hour:

- Suspicion of non-accidental injury.
- Post-traumatic seizure but no history of epilepsy.
- On initial emergency department assessment, GCS less than 14, or for children under one year GCS less than 15.
- At two hours after injury GCS less than 15.
- Suspected open or depressed skull fracture.
- Any sign of basal skull fracture.
- Focal neurological deficit.
- For children under one year, presence of bruise, swelling or laceration of more than 5cm on the head.

6.4.2 For children who have sustained a head injury and have more than one of the following risk factors, perform a CT head within one hour:

- Loss of consciousness lasting more than five minutes (witnessed).
- Abnormal drowsiness.
- Three or more discrete episodes of vomiting.
- Dangerous mechanism of injury (high-speed road traffic accident either as a pedestrian, cyclist or vehicle occupant, fall from a height of greater than 3m, high-speed injury from a projectile or other object).
- Amnesia.

6.4.3 Children who have sustained a head injury and only have one of the above risk factors (in 6.4.2) should be observed for a minimum of four hours after head injury. If during observation any of the risk factors below are identified, perform a CT head scan within one hour

- GCS less than 15.

- Further vomiting.
- A further episode of abnormal drowsiness.

6.5 Imaging the cervical spine

The cervical spine should be clinically assessed in all cases of head injury.

For children who have sustained a head injury, perform a CT cervical spine only if the following apply:

- GCS less than 13 on initial assessment.
- Patient has been intubated.
- Focal peripheral neurological signs.
- Paraesthesia in the upper or lower limbs.
- A definitive diagnosis of cervical spine injury is needed urgently (for example, before surgery).
- The patient is having other body areas scanned.
- There is a strong suspicion of injury despite normal x-rays.
- Plain x-rays are inadequate or show definite injury.

Children who are candidates for MRI should be discussed with the neuroradiologist on call.

7 Monitoring compliance with and the effectiveness of this document

- Audit
- Discussions in morbidity and mortality meetings
- Incident form reporting

8 References

Trauma East Manual of Procedures and Operations (TEMPO) 2014

<http://www.eotraumanetwork.nhs.uk/resources/trauma-east-manual-of-procedures-and-operations>

Head injury: Triage, assessment, investigation and early management of head injury in children, young people and adults NICE (CG 176) 2014

Schutzman SA et al. Evaluation and management of children younger than 2 years old with apparently minor head injury: proposed guidelines. Paediatrics: 107:5; May 2001: 983-993.

Dunning J, Batchelor J et al A meta-analysis of variables that predict significant intracranial injury in minor head trauma. Arch Dis Child: 89; 2004: 653-659.

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